BrightCart

BRIGHTCART

# Documentation Pack

5 documents

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AUTHOR BrightCart team

CONFIDENTIAL

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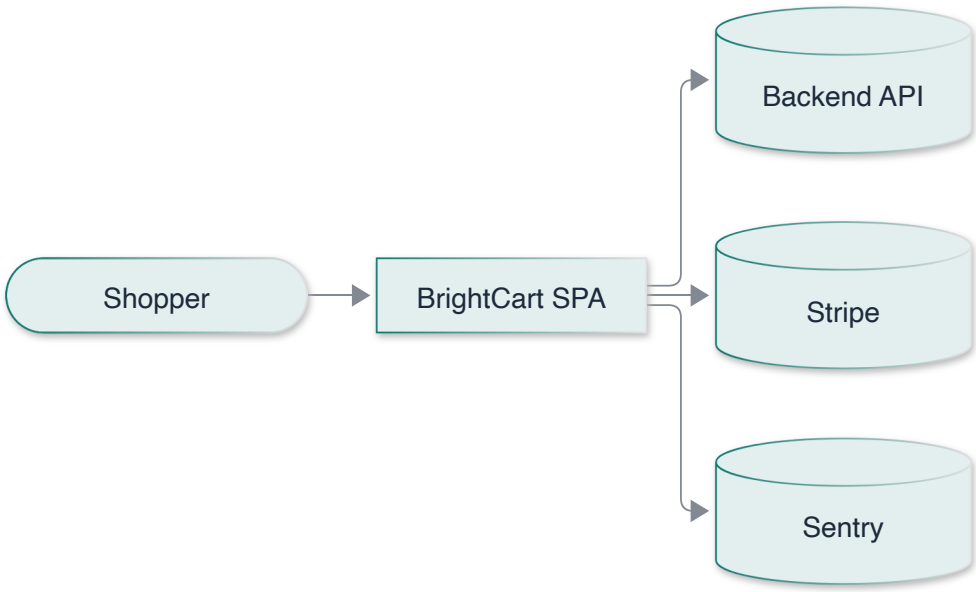
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# Architecture

## Overview

BrightCart is a client-side React single-page application. Vite builds it into static files that Vercel serves. The browser talks directly to three external systems: BrightCart's own backend API (`VITE_API_URL`), Stripe for payments and Sentry for error reporting. There is no server code in this repository.

## System context



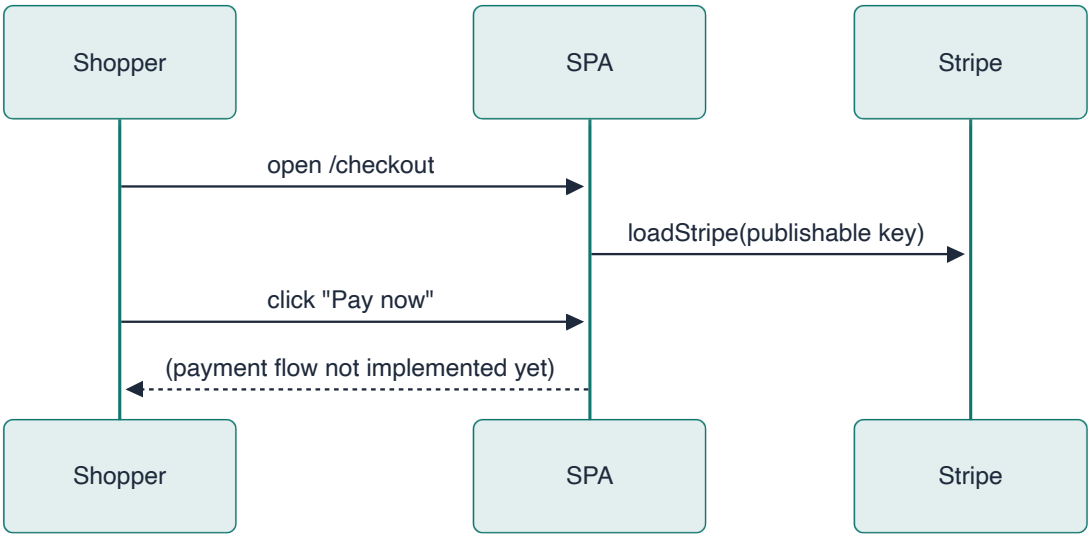

| External system | Used for | Evidence |
|---|---|---|
| Backend API | Product data (planned) | `VITE_API_URL` read in `src/pages/Home.tsx` |
| Stripe | Payments | `loadStripe` in `src/pages/Checkout.tsx` |
| Sentry | Error reporting | `Sentry.init` in `src/main.tsx` |
| Vercel | Hosting and deploys | `vercel.json`, `deploy` job in `.github/workflows/ci.yml` |

## Components

| Component | Responsibility | Tech | Location |
|---|---|---|---|
| App shell | Router and page layout | React Router 6 | `src/App.tsx` |
| Home page | Product listing | React, Tailwind | `src/pages/Home.tsx` |
| Checkout page | Starts payment | Stripe.js | `src/pages/Checkout.tsx` |

| Component | Responsibility | Tech | Location |
|---|---|---|---|
| ProductCard | Product tile | React | `src/components/ProductCard.tsx` |
| Bootstrap | Mounts React, starts Sentry | React DOM | `src/main.tsx` |

## Data flow: checkout




1. React Router renders `Checkout` for `/checkout` (`src/App.tsx`).
2. Importing the page calls `loadStripe` with `VITE_STRIPE_PUBLISHABLE_KEY`.
3. The **Pay now** button has no handler yet, so no payment is created ASSUMPTION that this is work in progress.

## Deployment view

GitHub Actions runs `lint` and `test` on every push and pull request. On `main` it also runs `npx vercel deploy --prod` with the `VERCEL_TOKEN` secret. `vercel.json` rewrites every path to `/`, so client-side routes survive a page refresh.

## Cross-cutting concerns

| Concern | Approach | Evidence |
|---|---|---|
| Configuration | Build-time `VITE_*` env vars | `.env.example` |
| Error handling | Sentry browser SDK | `src/main.tsx` |
| Styling | Tailwind with brand tokens | `tailwind.config.js` |
| Authentication | None in the frontend | no auth code found |

# Decision log

### ADR-001: Single-page app on Vite

- **Status:** Accepted
- **Context:** The storefront needs fast navigation and no server rendering ASSUMPTION.
- **Decision:** React 18 + Vite, routed client-side (`package.json`, `src/App.tsx`).
- **Consequences:** Static hosting is enough. SEO for product pages is limited without pre-rendering.

### ADR-002: Tailwind CSS for styling

- **Status:** Accepted
- **Context:** One consistent brand palette across components.
- **Decision:** Tailwind 3 with brand colours and fonts in `tailwind.config.js`.
- **Consequences:** No separate CSS files per component; styles live in class names.

### ADR-003: Vercel hosting

- **Status:** Accepted
- **Context:** Static SPA with preview needs ASSUMPTION.
- **Decision:** Deploy with the Vercel CLI from CI (`.github/workflows/ci.yml`).
- **Consequences:** Deploys depend on the `VERCEL_TOKEN` secret and Vercel availability.

# Related documents

Developer Guide · User Guide · CI/CD and Deployment

# Assumptions and open questions

| # | Item | Type | Why unverified | Who can confirm |
|---|---|---|---|---|
| A1 | Pay now is unfinished work, not a bug | assumption | button has no handler | frontend lead |
| A2 | Reasons behind ADR-001 and ADR-003 | assumption | inferred from code, no ADR files | tech lead |
| Q1 | What API does `VITE_API_URL` point to, and who owns it? | open question | backend not in repo | engineering lead |

# Budget and Resource Plan

*Planning estimates only, not financial advice.*

> **NOTE**
> These figures use **sample inputs** supplied to demonstrate the skill. They are not real BrightCart data.

The plan covers the people and tools needed to run the BrightCart storefront in 2027. Tooling lines match services found in the repo: Vercel (`vercel.json`), Sentry (`@sentry/react`) and GitHub Actions (`.github/workflows/ci.yml`).

## Inputs

Currency: **EUR**. Period: **12 months from 2027-01**, contingency **15%**.

| Input | Value | Source | Assumption |
|---|---|---|---|
| `currency` | EUR | user | — |
| `budget_start` | 2027-01 | user | — |
| `contingency_pct` | 15 | user | — |
| `team_roles` | 3 item(s) | user | — |
| `tooling_costs` | 3 item(s) | user | — |
| `fixed_infra_cost` | 120 | user | — |
| `one_time_costs` | 2 item(s) | user | — |

## Team

| Role | Allocation | Monthly rate | Monthly cost | Annual cost |
|---|---|---|---|---|
| Frontend developer | 100% | EUR 7,500 | EUR 7,500 | EUR 90,000 |
| Product designer | 40% | EUR 6,500 | EUR 2,600 | EUR 31,200 |
| Product manager | 20% | EUR 9,000 | EUR 1,800 | EUR 21,600 |

## Infrastructure and tooling

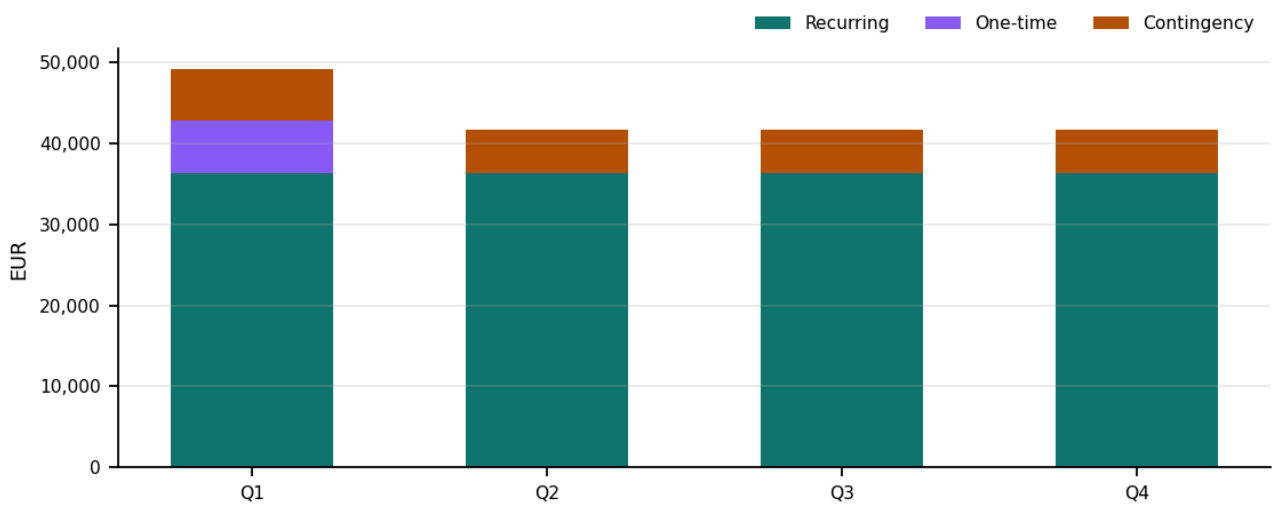
| Item | Monthly | Annual |
|---|---|---|
| Vercel Pro | EUR 20 | EUR 240 |

| Item | Monthly | Annual |
|---|---|---|
| Sentry Team | EUR 26 | EUR 312 |
| GitHub Team | EUR 16 | EUR 192 |
| Infrastructure (fixed) | EUR 120 | EUR 1,440 |

## Licences and one-time costs

| Item | Amount | Quarter |
|---|---|---|
| Accessibility audit | EUR 4,000 | Q1 |
| Product photography | EUR 2,500 | Q1 ASSUMPTION |

## Totals by quarter




*Budget by quarter (EUR)*

| Quarter | Recurring | One-time | Contingency | Total |
|---|---|---|---|---|
| Q1 (Jan 2027 – Mar 2027) | EUR 36,246 | EUR 6,500 | EUR 6,412 | EUR 49,158 |
| Q2 (Apr 2027 – Jun 2027) | EUR 36,246 | EUR 0 | EUR 5,437 | EUR 41,683 |
| Q3 (Jul 2027 – Sep 2027) | EUR 36,246 | EUR 0 | EUR 5,437 | EUR 41,683 |
| Q4 (Oct 2027 – Dec 2027) | EUR 36,246 | EUR 0 | EUR 5,437 | EUR 41,683 |
| **Year** | **EUR 144,984** | **EUR 6,500** | **EUR 22,723** | **EUR 174,207** |

## Formulas

```
team_monthly[r]   = allocation[r] × monthly_rate[r]
recurring_monthly = Σ team_monthly + Σ tooling + fixed_infra
recurring[q]      = 3 × recurring_monthly
contingency[q]    = (recurring[q] + one_time[q]) × contingency_pct
total[q]          = recurring[q] + one_time[q] + contingency[q]
```

One-time costs without a month are placed in Q1 ASSUMPTION.

## Related documents

1-Year Forecast · Stakeholders and RACI

## Assumptions and open questions

| # | Item | Type | Why unverified | Who can confirm |
|---|---|---|---|---|
| A1 | All rates and costs are sample values | assumption | demonstration run | finance |
| A2 | Product photography lands in Q1 | assumption | no month given | product owner |
| Q1 | Is the backend API ( `VITE_API_URL` ) budgeted elsewhere? | open question | backend not in this repo | engineering lead |

# Developer Guide

BrightCart is a React 18 single-page storefront built with Vite and styled with Tailwind CSS (`package.json`). This guide takes you from a fresh clone to a running dev server.

> **NOTE**
> Dependencies were not installed in the environment that generated this guide, so the npm commands below were read from `package.json` but not executed. They are marked `UNTESTED`.

## Prerequisites

| Tool | Version | Source |
|---|---|---|
| Node.js | 20 | `.nvmrc`, `node-version: 20` in `.github/workflows/ci.yml` |
| npm | bundled with Node 20 | CI runs `npm ci` |
| Git | any recent | — |

## Clone to running

1. Clone the repository and enter it. The remote URL is not recorded in the repo `ASSUMPTION`.

```
git clone <repository-url> brightcart
cd brightcart
```

2. Create your environment file and fill in the values:

```
cp .env.example .env
```

| Variable | Needed for |
|---|---|
| `VITE_API_URL` | Backend base URL used by `src/pages/Home.tsx` |
| `VITE_SENTRY_DSN` | Error reporting (`src/main.tsx`); empty disables it |
| `VITE_STRIPE_PUBLISHABLE_KEY` | Checkout (`src/pages/Checkout.tsx`) |

3. Install dependencies `UNTESTED`:

```
npm install
```

4. Start the dev server `UNTESTED`:

```
npm run dev
```

Vite serves the app on `http://localhost:5173` by default ASSUMPTION. No port is set in the repo.

## Project structure

```
brightcart/
├── index.html            # Vite entry; loads Poppins + Playfair Display
├── tailwind.config.js    # brand colours and font families
├── vercel.json           # SPA rewrite: every path serves /
├── public/               # logo.svg, favicon.svg
├── src/
│   ├── main.tsx          # React root + Sentry.init
│   ├── App.tsx           # routes: / and /checkout
│   ├── index.css         # Tailwind layers + CSS variables
│   ├── pages/            # Home.tsx, Checkout.tsx
│   ├── components/       # ProductCard.tsx
│   └── App.test.tsx      # Vitest + Testing Library
└── .github/workflows/ci.yml
```

## Common workflows

| Command | What it does | Tested |
|---|---|---|
| `npm run dev` | Start the Vite dev server | UNTESTED |
| `npm run build` | Type-check ( `tsc -b` ) and build to `dist/` | UNTESTED |
| `npm run preview` | Serve the production build locally | UNTESTED |
| `npm run lint` | ESLint over the repo | UNTESTED |
| `npm test` | Run Vitest once ( `vitest run` ) | UNTESTED |

### Add a page

1. Create `src/pages/<Name>.tsx`, following `src/pages/Home.tsx`.
2. Register a route in `src/App.tsx`:

```
<Route path="/orders" element={<Orders />} />
```

3. Production needs no server change: `vercel.json` already rewrites every path to `/`.

### Use the brand styles

Use the Tailwind tokens from `tailwind.config.js` rather than raw hex values: `bg-brand`, `text-brand-700`, `font-display` (Playfair Display) and `font-sans` (Poppins).

# Testing

```
npm test
```

`src/App.test.tsx` has 2 tests: the home page renders, and a placeholder for the checkout route. See Testing Strategy.

# Debugging tips

- Vite only exposes variables prefixed with `VITE_` to the browser. A missing value reads as `undefined` rather than throwing, so check `.env` first when a feature silently fails.
- Sentry starts in `src/main.tsx` with `VITE_SENTRY_DSN`. Leave it empty locally to avoid sending development errors.
- `loadStripe` in `src/pages/Checkout.tsx` runs at module load. An invalid publishable key shows up in the browser console as soon as the checkout page is imported.

# Related documents

Architecture · User Guide · Testing Strategy

# Assumptions and open questions

| # | Item | Type | Why unverified | Who can confirm |
|---|---|---|---|---|
| A1 | Repository URL | assumption | no git remote in the repo | maintainer |
| A2 | Dev server port 5173 | assumption | Vite default, not configured | frontend lead |
| A3 | npm commands work as listed | assumption | dependencies not installed during generation | frontend lead |
| Q1 | Where does the backend behind `VITE_API_URL` live? | open question | not in this repo | engineering lead |

# 1-Year Forecast

*Planning estimates only, not financial advice.*

> **NOTE**
> These figures use **sample inputs** supplied to demonstrate the skill. They are not real BrightCart data.

This forecast covers active shoppers, Vercel/hosting cost and usage-based revenue for 2027. Hosting is Vercel (`vercel.json`, deploy job in `.github/workflows/ci.yml`); the only paid runtime services visible in the code are Sentry (`src/main.tsx`) and Stripe (`src/pages/Checkout.tsx`).

## Assumptions

Currency: **EUR**. Period: **monthly, Jan 2027 – Dec 2027**.

| Input | Value | Source | Assumption |
|---|---|---|---|
| `currency` | EUR | user | — |
| `forecast_start` | 2027-01 | user | — |
| `starting_users` | 2500 | user | — |
| `monthly_growth_pct` | 3-9 | user | — |
| `infra_cost_per_user` | 0.02 | user | — |
| `fixed_infra_cost` | 120 | user | — |
| `revenue_model` | usage-based | user | — |
| `arpu` | 4 | user | — |
| `paid_conversion_pct` | 2-5 | user | — |

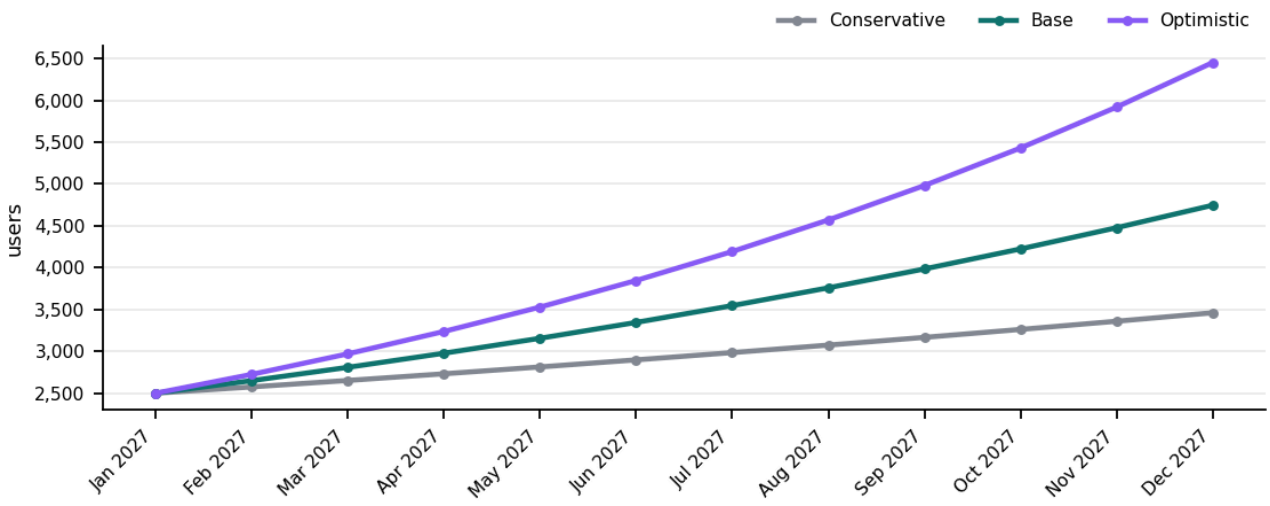
## Method

```
users[m]   = starting_users × (1 + growth)^m          (m = 0..11)
infra[m]   = fixed_infra_cost + users[m] × infra_cost_per_user
paying[m]  = users[m] × paid_conversion
revenue[m] = paying[m] × arpu
net        = Σ revenue - Σ infra
```

## Scenarios

| Scenario | Growth / month | Conversion |
|---|---|---|
| Conservative | 3% | 2% |
| Base | 6% | 3.5% |
| Optimistic | 9% | 5% |

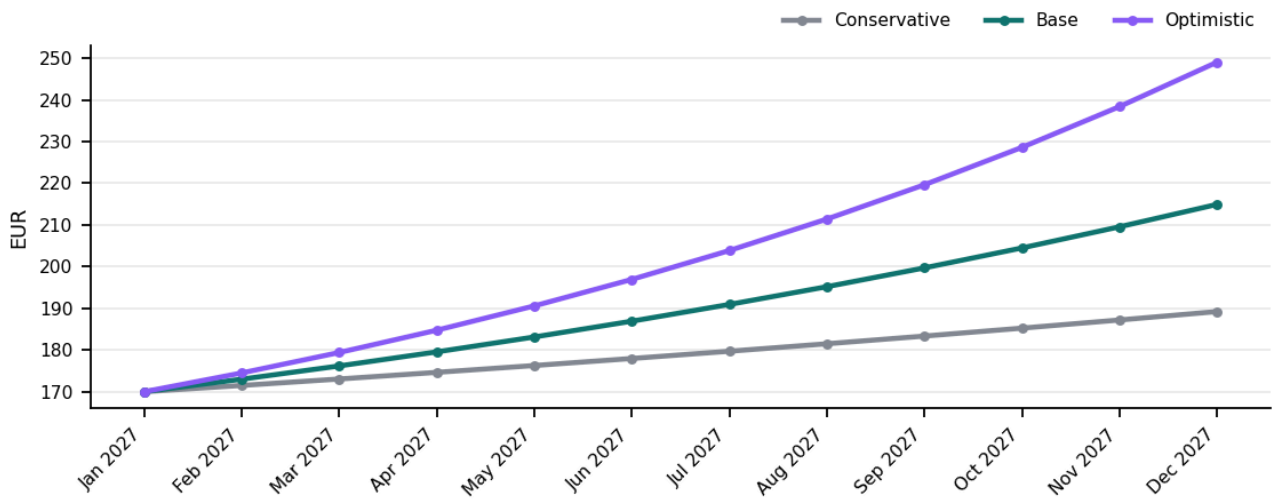
## Users and usage




*Active users by scenario*

| Month | Conservative | Base | Optimistic |
|---|---|---|---|
| Jan 2027 | 2,500 | 2,500 | 2,500 |
| Feb 2027 | 2,575 | 2,650 | 2,725 |
| Mar 2027 | 2,652 | 2,809 | 2,970 |
| Apr 2027 | 2,732 | 2,978 | 3,238 |
| May 2027 | 2,814 | 3,156 | 3,529 |
| Jun 2027 | 2,898 | 3,346 | 3,847 |
| Jul 2027 | 2,985 | 3,546 | 4,193 |
| Aug 2027 | 3,075 | 3,759 | 4,570 |
| Sep 2027 | 3,167 | 3,985 | 4,981 |
| Oct 2027 | 3,262 | 4,224 | 5,430 |
| Nov 2027 | 3,360 | 4,477 | 5,918 |
| Dec 2027 | 3,461 | 4,746 | 6,451 |

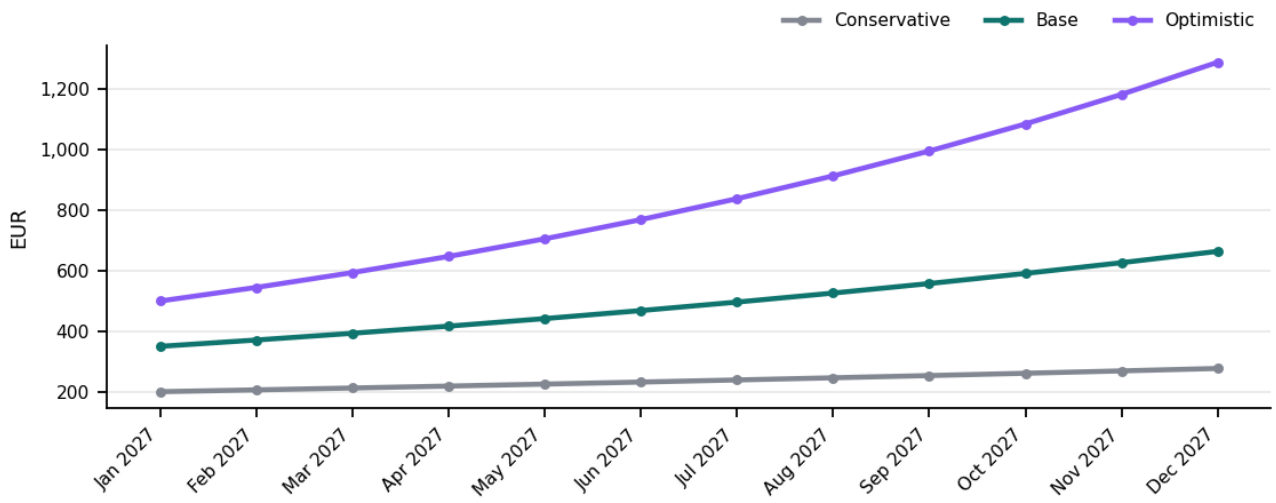
# Infrastructure cost




*Monthly infrastructure cost (EUR)*

| Month | Conservative | Base | Optimistic |
|---|---|---|---|
| Jan 2027 | EUR 170 | EUR 170 | EUR 170 |
| Feb 2027 | EUR 172 | EUR 173 | EUR 174 |
| Mar 2027 | EUR 173 | EUR 176 | EUR 179 |
| Apr 2027 | EUR 175 | EUR 180 | EUR 185 |
| May 2027 | EUR 176 | EUR 183 | EUR 191 |
| Jun 2027 | EUR 178 | EUR 187 | EUR 197 |
| Jul 2027 | EUR 180 | EUR 191 | EUR 204 |
| Aug 2027 | EUR 181 | EUR 195 | EUR 211 |
| Sep 2027 | EUR 183 | EUR 200 | EUR 220 |
| Oct 2027 | EUR 185 | EUR 204 | EUR 229 |
| Nov 2027 | EUR 187 | EUR 210 | EUR 238 |
| Dec 2027 | EUR 189 | EUR 215 | EUR 249 |

# Revenue




*Monthly revenue (EUR)*

| Month | Conservative | Base | Optimistic |
|---|---|---|---|
| Jan 2027 | EUR 200 | EUR 350 | EUR 500 |
| Feb 2027 | EUR 206 | EUR 371 | EUR 545 |
| Mar 2027 | EUR 212 | EUR 393 | EUR 594 |
| Apr 2027 | EUR 219 | EUR 417 | EUR 648 |
| May 2027 | EUR 225 | EUR 442 | EUR 706 |
| Jun 2027 | EUR 232 | EUR 468 | EUR 769 |
| Jul 2027 | EUR 239 | EUR 496 | EUR 839 |
| Aug 2027 | EUR 246 | EUR 526 | EUR 914 |
| Sep 2027 | EUR 253 | EUR 558 | EUR 996 |
| Oct 2027 | EUR 261 | EUR 591 | EUR 1,086 |
| Nov 2027 | EUR 269 | EUR 627 | EUR 1,184 |
| Dec 2027 | EUR 277 | EUR 664 | EUR 1,290 |

# Summary

| Metric (year 1) | Conservative | Base | Optimistic | Range |
|---|---|---|---|---|
| Active users at month 12 | 3,461 | 4,746 | 6,451 | 3,461 – 6,451 |
| Total infra cost | EUR 2,150 | EUR 2,283 | EUR 2,447 | EUR 2,150 – EUR 2,447 |
| Total revenue | EUR 2,838 | EUR 5,904 | EUR 10,070 | EUR 2,838 – EUR 10,070 |
| Net (revenue – infra) | EUR 689 | EUR 3,621 | EUR 7,623 | EUR 689 – EUR 7,623 |

## What would change this forecast

- Each extra percentage point of monthly growth adds about **516** users by month 12 (base scenario).
- Infrastructure cost scales linearly with users; a cheaper tier or caching lowers `infra_cost_per_user`.

## Related documents

Budget and Resource Plan · Roadmap

## Assumptions and open questions

| # | Item | Type | Why unverified | Who can confirm |
|---|---|---|---|---|
| A1 | All forecast inputs are sample values | assumption | demonstration run | product owner |
| A2 | Stripe fees are not modelled as infra cost | assumption | no pricing data in repo | finance |
| Q1 | What does "active user" mean: shopper or maker? | open question | not defined in code | product owner |

# User Guide

BrightCart is an online shop for independent makers. This guide shows shoppers how to find a product and pay for it.

## Getting started

Open BrightCart in any modern browser. The production address isn't recorded in the repository ASSUMPTION. You don't need an account to browse.

## Tasks

### Browse products

1. Open the home page. The heading reads **Fresh from the makers**.
2. Each product appears as a card with its name and price, for example **Ceramic mug, $24**.

> **TIP**
> Product cards show the price in dollars ( `src/components/ProductCard.tsx` ).

### Pay for your order

1. Go to the **checkout** page ( `/checkout` ).
2. Select **Pay now**.

> **WARNING**
> In this version, **Pay now** does not start a payment yet. The button has no action attached ( `src/pages/Checkout.tsx` ), so no money is taken.

## Account and settings

BrightCart has no sign-in, profile or settings pages in this version.

## Troubleshooting

| Problem | What to do |
|---|---|
| The page is blank after refreshing | Reload the home page. The site serves every address from the same page ( `vercel.json` ) |
| Checkout shows an error in the browser console | The payment key is missing; contact the BrightCart team |

## Related documents

Quick Start · Troubleshooting and FAQ

## Assumptions and open questions

| # | Item | Type | Why unverified | Who can confirm |
|---|---|---|---|---|
| A1 | Production URL | assumption | not in repo | product owner |
| Q1 | When will Pay now take payments? | open question | handler not implemented | product owner |
| Q2 | Will shoppers get accounts or order history? | open question | no such pages in repo | product owner |